

Assessment #1 - Current State of the Printed Circuit Board Industry

Sai Dandamudi

Lesli Rusell

ISM 1

29 August 2026

PCB Industry

Works Cited

- Atchison, Ian. "Advanced PCB Materials: Beyond FR-4 for High Performance." Bencor, 3 Mar. 2025, <https://bencor-llc.com/beyond-fr-4-exploring-advanced-materials-for-high-performance-pcbs/>. Bishop, Steve X.
- "5 Historical Events That Shaped PCBs for Electronic Manufacturers." Electronic Manufacturer - Levison Enterprises - US Based Electronic Manufacturer, 8 Mar. 2018, <https://levisonenterprises.com/5-events-pcb-history/>. Kong, Kyle.
- "Industry Trends Electronics/ICT August 2026." Atradius, 5 Aug. 2026, <https://atradius.us/knowledge-and-research/reports/industry-trends-electronics-ict-august-2026>. Research and Markets.
- "United States Printed Circuit Board Market Report 2025-2033 by Type, Substrate, End Use, States and Company Analysis." Yahoo Finance, Yahoo! Finance, 10 Nov. 2025, https://finance.yahoo.com/news/united-states-printed-circuit-board-162800910.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAH0ie8ICSFFdkR5oHDW091qSvWK1wjuhO3Pl3aTo6YRTybay9cbPCy_fyyT3

ZUwHjTlruM6SS19VYEZUe1S7ixYzMmaqE9Ohp5-40mnivykmzQCesN_B4SOUiF96
 p32drXQ7ChmiMU0zDpg_yFNfHiEbDFTRICpeXA_RjPHR6Yl6.Sushmitha V. “How’s
 AI Transforming the Circuit Board Industry?” Sierra Circuits, 14 Jan. 2026,
<https://www.protoexpress.com/blog/how-s-ai-transforming-circuit-board-industry/>.

PCBs, or printed circuit boards, are wafers of silicon with copper track in between then allowing for the creation of highly efficient, cheap, and small circuits. PCBs are used everywhere from computers to cars to medical equipment. Anything which needs to process data probably has a PCB behind it. This background of necessity raises the question of what the PCB industry is like at the moment.

What’s Happening Right Now?

Atradius states “We expect global electronics/ICT production to grow by 11% in 2026 and by 7.3% in 2027 after a 10% increase in 2025.” (Atradius) and they cite this trend to the development of Artificial Intelligence (AI) in recent years. AI datacenters are the main drivers of this trend. As it has been stated before PCBs are used in almost every electronic product, therefore as the electronics market grows so will the PCB industry. To support this even further, Yahoo Finance in their market report for 2025-2033 say that the PCB market will grow by 10.62 billion dollars in the timespan (Yahoo Finance). They associate the growth of the PCB industry with again the rise in AI as well as increased investments in 5G and renewable energy. This is significant as it supports the idea that even though AI is a significant contributor they aren't the sole reason the industry is expanding.

What's Disrupting the Field?

PCBs being such an essential field means that trillions of dollars of research and development is poured into it. Currently most PCBs use Flame Retardant - 4, a glass-reinforced epoxy laminate that has its limitations. Bencor states improvements such as improved high-frequency performance, thermal conductivity, heat dissipation, temperature resistance, dielectric stability, dimensional stability, and overall reliability have been achieved (Bencor). Using this we can go on to infer that PCBs will continue to evolve as more materials are discovered. In addition AI has been shown to be used to design more efficient PCBs by performing analysis on created designs to look for improvements (Sierra Circuits). This creates both excitement in the sense that PCBs are evolving in terms of material but also the question of whether PCB designers are no longer relevant.

Old Ways vs. New Ways

PCBs have changed in many ways from material to use but the most important change was in production. According to Levison Enterprises, in the past PCBs had to be made through a labor intensive process namely through-hole construction where wires were individually soldered into place. While this was adequate way back when, the demand for PCBs has only risen (Kong). Therefore Moe Abramson and Stanislaus Danko developed an automated assembly process using printed wiring patterns and dip soldering techniques to speed up the process, which it did. This change had also accelerated the growth of the industry as well by making PCB manufacture more accessible.

Looking Ahead & Keeping Your Research Fresh

The PCB field is vital and will continue to grow as long as technology continues to be in demand for its speed and accuracy. To support its permanence, the industry isn't reliant on AI alone but a multitude of fields, which means it will continue to grow regardless of the growth of AI. There have been vast advancements in PCB material and design, mirroring the past advancement in manufacturing from history with AI helping with design. To sum it all up, the PCB industry is going to continue to grow and be in high demand.